

Demonstration of Proposed Features

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document presents the features proposed during the planning phase of the **Voice-Based Concept Understanding Analyser** and their implementation status. It demonstrates that the proposed AI-powered solution for evaluating conceptual understanding through voice and text input has been successfully developed and tested.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrate d (Yes / No)	Remarks
1	Home Page	Landing page introducing the Voice-Based Concept Understanding Analyser and providing navigation to different modules.	Implemented	Yes	User interface and navigation verified successfully.
2	About Page	Provides project objectives, workflow, technologies used, and benefits of AI-based concept evaluation	Implemented	Yes	Displays project information correctly.
3	Voice & Text Input Interface	Allows users to explain concepts through voice recording or manual text input for evaluation.	Implemented	Yes	Successfully tested with multiple concept explanations.
4	Speech-to-Text Conversion	Converts spoken explanations into text using the browser's Web Speech API before NLP processing.	Implemented	Yes	Accurate speech recognition verified during demonstration.
5	NLP-Based Concept Analysis	Performs semantic similarity analysis, identifies missing concepts and misconceptions, and evaluates conceptual understanding	Implemented	Yes	Successfully generated mastery scores and concept analysis results.
6	Personalized Feedback & Mastery Score	Generates concept coverage, mastery score, personalized feedback, and improvement suggestions based on the evaluation.	Implemented	Yes	Feedback generation validated with different user responses.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%